

CSE 3031 Computer Architecture

Homework 3

Spring 2026

Deadline: Midnight of Thursday, Apr. 2

Problem 1 (40 pts, 10/10/10/10): Consider a 64-byte cache with 8 byte blocks, an associativity of 2 and LRU block replacement. Virtual addresses are 16 bits. The cache is physically tagged. The processor has 16KB of physical memory (Hint: Calculate the physical address size based on the physical memory size).

- [10] What is the total number of tag bits?
- [10] If the cache is virtually indexed, what is the minimum page size to avoid the synonym problem?
- [10] Assume each page is 64 bytes. How large would a page table be given that each page requires 4 protection bits in addition to the physical page number, and entries must be an integral number of bytes?
- [10] Why are the first-level caches usually split (instructions and data are in different caches) while the L2 cache is usually unified (instructions and data are both in the same cache)?

Problem 2 (20 pts) Consider the code snippet below. Suppose it is executed on a system with a 2-way set associative 16KB data cache with 32-byte blocks, 32-bit words, and an LRU replacement policy. Assume that `int` is word-sized. Also assume that the address of `a` is 0x0, that `i` and `x` are in registers, and that the cache is initially empty. How many data cache misses are there? How many hits are there? Please justify your answer.

```
int i;
int a[1024*1024];
int x = 0;
for(i = 0; i < 1024; i++)
{
    x += a[i] + a[1024*i];
}
```

Problem 3 (15 pts, 5/10): Whereas larger caches have lower miss rates, they also tend to have longer hit times. Assume a direct-mapped 8 KB cache has 0.22 ns hit time and a miss rate of m_1 ; also assume a 4-way set associative 64 KB cache has 0.52 ns hit time and a miss rate of m_2 .

- [5] If the miss penalty is 100 ns, when would it be advantageous to use the smaller direct-mapped cache to reduce the overall memory access time?
- [10] Repeat part (a) for miss penalties of 10 ns and 1,000 ns. Conclude when it may be advantageous to use a smaller direct-mapped cache.

Problem 4 (15 pts): A program is running on a computer with a four-entry fully associative TLB cache (Table 3). Tables 2 and 3 show the page table contents and a trace of virtual page numbers accessed by a program, respectively.

For each access in Table 3, indicate whether it produces a TLB hit or miss and, if it accesses the page table, whether it produces a page hit or fault. Put an X under the page table column if it is not accessed. Put an X under the page table column if the page table is not accessed for the address translation.

Table 1: TLB contents

Virtual page index	Physical page number	Entry valid
5	30	1
7	1	0
10	10	1
15	25	1

Table 2: Page table contents

Virtual page index	Physical page number	Entry valid
0	3	Y
1	7	N
2	6	N
3	5	Y
4	14	Y
5	39	Y
6	26	Y
7	11	Y
8	13	N
9	18	N
10	10	Y
11	56	Y
12	110	Y
13	33	Y
14	12	N
15	25	Y

Table 3: Page access trace

Virtual page accessed	TLB (hit or miss)	Page table (hit or fault or X)
1		
5		
9		
14		
10		
6		
15		
12		
7		
2		

Problem 5 (20 pts, 10/10): A computer has 16 pages of virtual address space but only four physical page frames. Initially, the physical memory is empty. A program references the virtual pages in the order: 0 2 4 5 2 4 3 11 2 10.

- [10] Which references cause a page fault with the LRU page replacement policy? Indicate the virtual pages in the physical memory after all references.
- [10] Which references cause a page fault with the FIFO page replacement policy? Indicate the virtual pages in the physical memory after all references.

How to submit: Write your answer in a docx or pdf file. Upload the file through the submission link on Canvas.